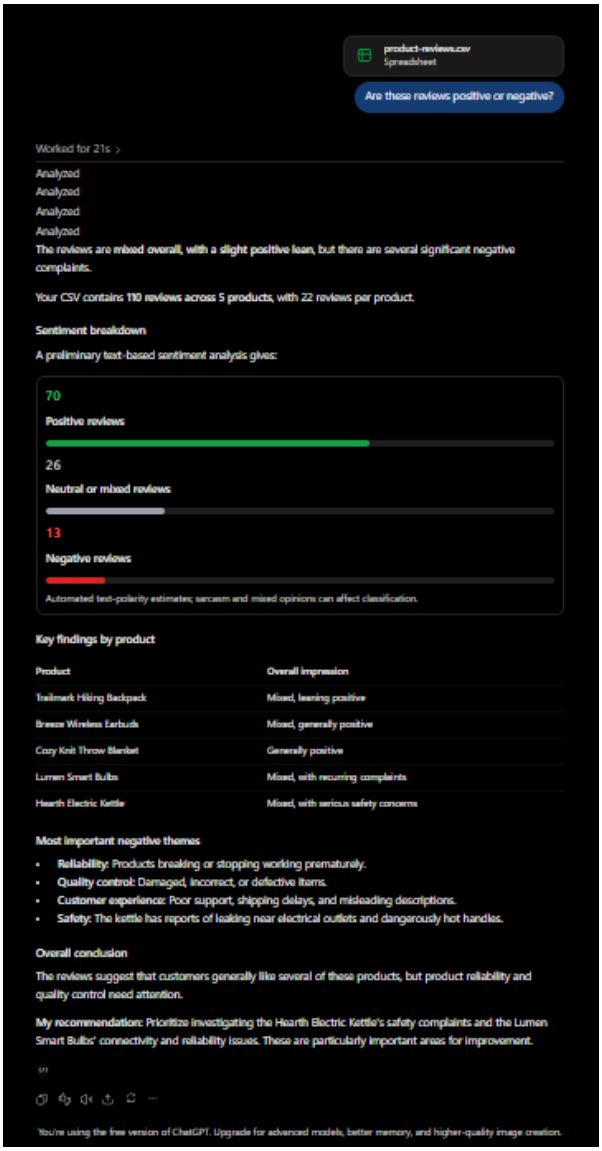
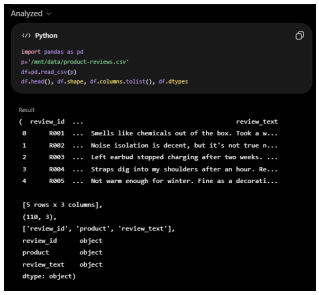
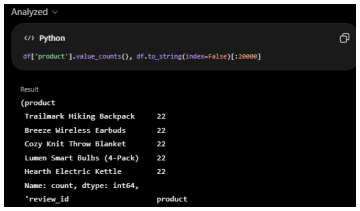
Kevin Hernandez
2026 November 2

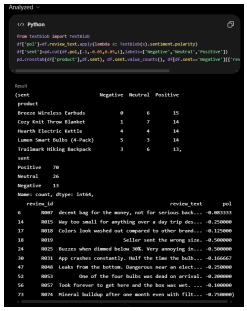
Prompt Engineering Assignment 2

Part 1: Start Simple. Asking ChatGPT to analyze the document, we inputted “Are these reviews positive or negative?”. Prompt Version 1







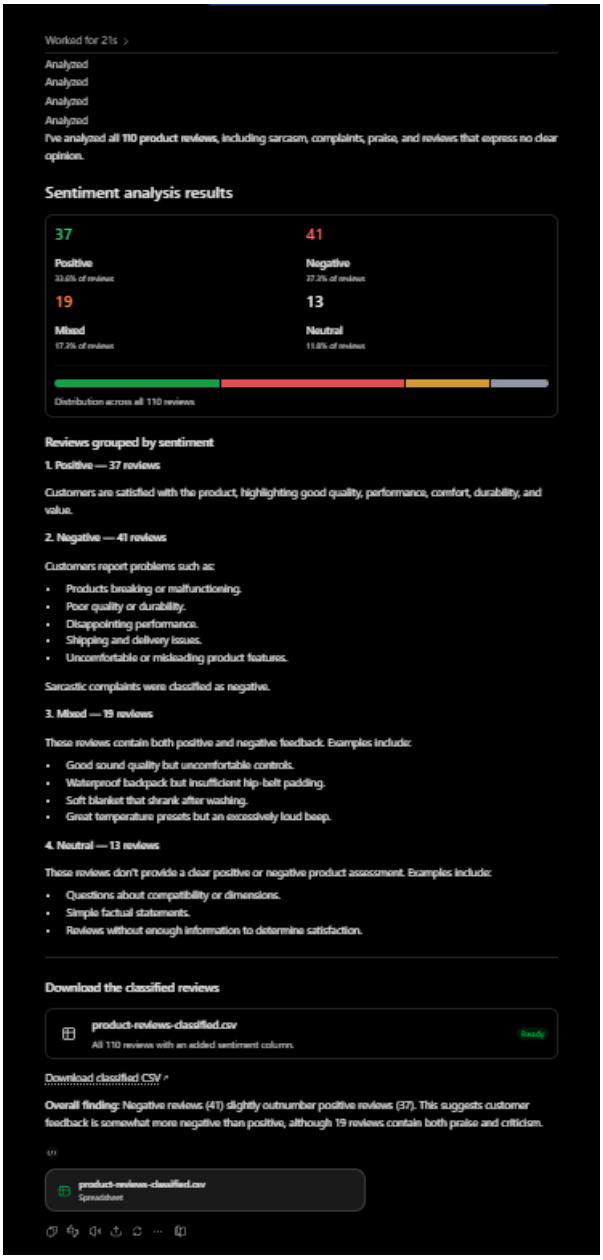


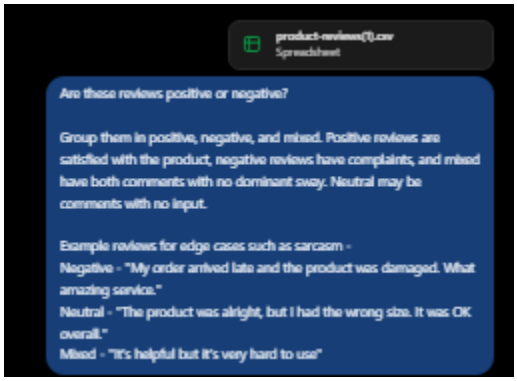


As we can see from the prompt, the model actually attempted to mark each individual review through a sentiment analysis tool, and returned the output. It summarized the results of the tool, and attempted to sort them by negative, neutral, or positive. It did not however create a separate file detailing the findings or sorting/counting them, only reporting the output. For mixed reviews it labeled them as neutral.

Part 2: Improved prompts with techniques

Version 2 - Defining Categories + Few-shot Prompting





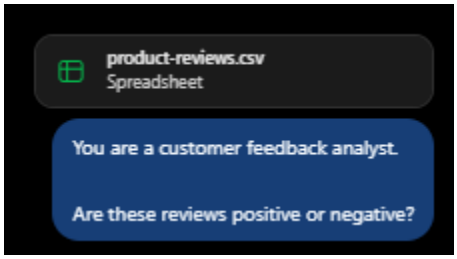
Distribution is a LOT different. 70 positive, 26 neutral, 13 negative shifted to 37 positive, 41 negative, 19 mixed, and 13 neutral. The model also provided a document detailing each categorization.

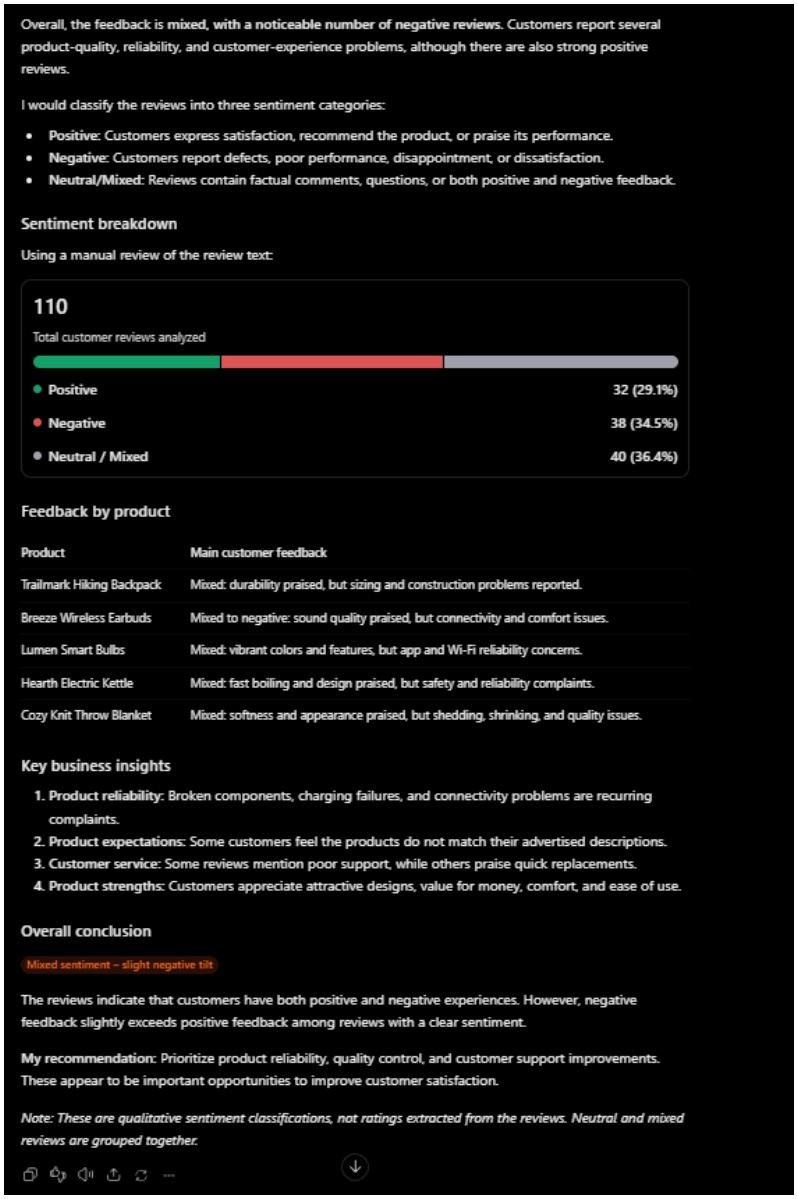
Version 3 - Output format + Constraints



The model added upon its reasoning, but given our specific constraints (and oversight in the formatting request) it ended up not tallying the reviews, but it did provide rationale for each.

Version 4 - Role Prompting





Overall, the feedback is **mixed, with a noticeable number of negative reviews**. Customers report several product-quality, reliability, and customer-experience problems, although there are also strong positive reviews.

I would classify the reviews into three sentiment categories:

- **Positive:** Customers express satisfaction, recommend the product, or praise its performance.
- **Negative:** Customers report defects, poor performance, disappointment, or dissatisfaction.
- **Neutral/Mixed:** Reviews contain factual comments, questions, or both positive and negative feedback.

Sentiment breakdown

Using a manual review of the review text:

110
Total customer reviews analyzed

Positive	32 (29.1%)
Negative	38 (34.5%)
Neutral / Mixed	40 (36.4%)

Feedback by product

Product	Main customer feedback
Trailmark Hiking Backpack	Mixed: durability praised, but sizing and construction problems reported.
Breeze Wireless Earbuds	Mixed to negative: sound quality praised, but connectivity and comfort issues.
Lumen Smart Bulbs	Mixed: vibrant colors and features, but app and Wi-Fi reliability concerns.
Hearth Electric Kettle	Mixed: fast boiling and design praised, but safety and reliability complaints.
Cozy Knit Throw Blanket	Mixed: softness and appearance praised, but shedding, shrinking, and quality issues.

Key business insights

1. **Product reliability:** Broken components, charging failures, and connectivity problems are recurring complaints.
2. **Product expectations:** Some customers feel the products do not match their advertised descriptions.
3. **Customer service:** Some reviews mention poor support, while others praise quick replacements.
4. **Product strengths:** Customers appreciate attractive designs, value for money, comfort, and ease of use.

Overall conclusion

Mixed sentiment – slight negative tilt

The reviews indicate that customers have both positive and negative experiences. However, negative feedback slightly exceeds positive feedback among reviews with a clear sentiment.

My recommendation: Prioritize product reliability, quality control, and customer support improvements. These appear to be important opportunities to improve customer satisfaction.

Note: These are qualitative sentiment classifications, not ratings extracted from the reviews. Neutral and mixed reviews are grouped together.

More detailed than simply rating them, including implicit justification given the role.

Part 3: Reviewing the Model's Work

From these 10 samples, these are my manual reviews.

R092	Cozy Knit Throw	It's a blanket. Keeps me warm.				Neutral
R093	Cozy Knit Throw	Delivery driver left it in the rain.				Negative
R094	Breeze Wireless	Bought these as a gift for my brother. He hasn't opened them yet.				Neutral
R095	Trailmark Hiking	The stitching is already coming apart and I've only used it twice.				Negative
R096	Lumen Smart Bu	Requires an account and wants my location data just to turn on a light. No thanks.				Negative
R097	Breeze Wireless	Love the sound, hate the touch controls. Every time I adjust them the music pauses.				Mixed
R098	Lumen Smart Bu	Nothing says "smart home" like having to unplug the router to turn on my lights.				Negative
R099	Cozy Knit Throw	The knit pattern snags on everything. Pretty but impractical.				Mixed
R100	Lumen Smart Bu	Stopped working entirely after a month. Waste of money.				Negative
R101	Breeze Wireless	Does anyone know if these work with a PS5?				Neutral
R102	Cozy Knit Throw	Best throw blanket I've owned. Heavy, warm, and soft.				Positive

We shall be comparing the model's response to Version 2, where we gave more specific categorizations and examples. It actually ended up lining up quite well - all were the same.

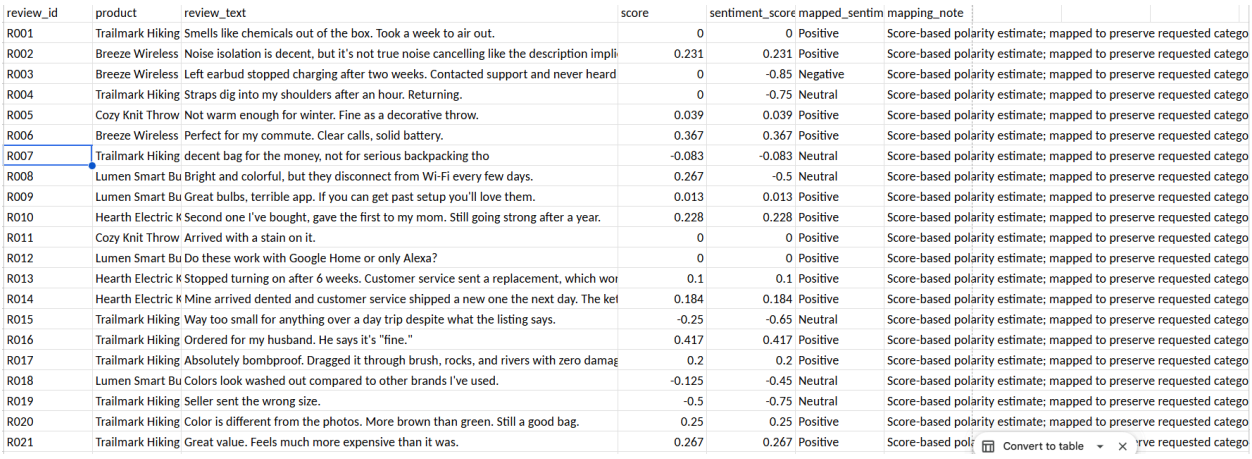
93	R092	Cozy Knit T	It's a blank	Neutral
94	R093	Cozy Knit T	Delivery dr	Negative
95	R094	Breeze Wir	Bought the	Neutral
96	R095	Trailmark H	The stitchir	Negative
97	R096	Lumen Sma	Requires ar	Negative
98	R097	Breeze Wir	Love the sc	Mixed
99	R098	Lumen Sma	Nothing sa	Negative
100	R099	Cozy Knit T	The knit pa	Mixed
101	R100	Lumen Sma	Stopped w	Negative
102	R101	Breeze Wir	Does anyor	Neutral
103	R102	Cozy Knit T	Best throw	Positive

Analysis

For **version 2**, I used defined categories and few-shot prompting by clarifying what each review should be grouped under (Mixed/Neutral/Positive/Negative) and gave an example of each, so the model had a clearer idea of what to categorize each. **Version 3** I utilized output format and constraints, and these worked just as well. Having the model work and explain the answer as it was scoring enabled it to think through and label each response similarly well. **Version 4**, which was solely role-prompting, also provided adequate details that provided enough information for its use case

Compared to **Version 1**, which used Sentiment Analysis (dove under the hood in a follow-up prompt) - all other responses performed much better. Version 1 had scored 70 positive reviews, but the sentiment analysis labeled sarcasm and other similar gray area/sarcastic reviews as

positive. It did not originally provide an Excel sheet, and when I tried to look under the hood of its analysis score, it tried to formulate a new answer with a more normal distribution like the other versions instead of expanding upon the original.



review_id	product	review_text	score	sentiment_score	mapped_sentim	mapping_note
R001	Trailmark Hiking	Smells like chemicals out of the box. Took a week to air out.	0	0	Positive	Score-based polarity estimate; mapped to preserve requested catego
R002	Breeze Wireless	Noise isolation is decent, but it's not true noise cancelling like the description impli	0.231	0.231	Positive	Score-based polarity estimate; mapped to preserve requested catego
R003	Breeze Wireless	Left earbud stopped charging after two weeks. Contacted support and never heard	0	-0.85	Negative	Score-based polarity estimate; mapped to preserve requested catego
R004	Trailmark Hiking	Straps dig into my shoulders after an hour. Returning.	0	-0.75	Neutral	Score-based polarity estimate; mapped to preserve requested catego
R005	Cozy Knit Throw	Not warm enough for winter. Fine as a decorative throw.	0.039	0.039	Positive	Score-based polarity estimate; mapped to preserve requested catego
R006	Breeze Wireless	Perfect for my commute. Clear calls, solid battery.	0.367	0.367	Positive	Score-based polarity estimate; mapped to preserve requested catego
R007	Trailmark Hiking	decent bag for the money, not for serious backpacking tho	-0.083	-0.083	Neutral	Score-based polarity estimate; mapped to preserve requested catego
R008	Lumen Smart Bu	Bright and colorful, but they disconnect from Wi-Fi every few days.	0.267	-0.5	Neutral	Score-based polarity estimate; mapped to preserve requested catego
R009	Lumen Smart Bu	Great bulbs, terrible app. If you can get past setup you'll love them.	0.013	0.013	Positive	Score-based polarity estimate; mapped to preserve requested catego
R010	Hearth Electric K	Second one I've bought, gave the first to my mom. Still going strong after a year.	0.228	0.228	Positive	Score-based polarity estimate; mapped to preserve requested catego
R011	Cozy Knit Throw	Arrived with a stain on it.	0	0	Positive	Score-based polarity estimate; mapped to preserve requested catego
R012	Lumen Smart Bu	Do these work with Google Home or only Alexa?	0	0	Positive	Score-based polarity estimate; mapped to preserve requested catego
R013	Hearth Electric K	Stopped turning on after 6 weeks. Customer service sent a replacement, which wor	0.1	0.1	Positive	Score-based polarity estimate; mapped to preserve requested catego
R014	Hearth Electric K	Mine arrived dented and customer service shipped a new one the next day. The ket	0.184	0.184	Positive	Score-based polarity estimate; mapped to preserve requested catego
R015	Trailmark Hiking	Way too small for anything over a day trip despite what the listing says.	-0.25	-0.65	Neutral	Score-based polarity estimate; mapped to preserve requested catego
R016	Trailmark Hiking	Ordered for my husband. He says it's "fine."	0.417	0.417	Positive	Score-based polarity estimate; mapped to preserve requested catego
R017	Trailmark Hiking	Absolutely bombproof. Dragged it through brush, rocks, and rivers with zero damag	0.2	0.2	Positive	Score-based polarity estimate; mapped to preserve requested catego
R018	Lumen Smart Bu	Colors look washed out compared to other brands I've used.	-0.125	-0.45	Neutral	Score-based polarity estimate; mapped to preserve requested catego
R019	Trailmark Hiking	Seller sent the wrong size.	-0.5	-0.75	Neutral	Score-based polarity estimate; mapped to preserve requested catego
R020	Trailmark Hiking	Color is different from the photos. More brown than green. Still a good bag.	0.25	0.25	Positive	Score-based polarity estimate; mapped to preserve requested catego
R021	Trailmark Hiking	Great value. Feels much more expensive than it was.	0.267	0.267	Positive	Score-based pol

Convert to table

Of course, this may also be a hallucination as it tries to replicate the ‘original’ response. This presents some of the risks when inspecting at face value - several of the sarcastic or neutral comments pass as positive. The biggest difference in prompting would have been providing examples and reasoning. By providing examples, it made the process much simpler for the model to understand the scoring behind each review, not solely based on sentiment scoring or which tool it tried to utilize. Besides that, the model did all jobs well, scoring each review, but some prompts it did not provide an answer which made it simpler for future responses to note that it should format in a specific category.

Takeaway

Formatting and reasoning techniques for the model are important to use to get the most use. The first attempt utilised a tool to analyze the spreadsheets, which provided a skewed and incorrect assessment for our purposes. By providing examples, context, a reasoning system, and more rigid constraints, it guided the model much clearer in how to score the reviews. This experiment best portrayed how to get the most out of prompting to avoid errors.